Header Footer Antigravity

Project Category: B1AI DEVOP1
Author: Roni Fernandes Dias / François LANGE
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PDF Header and Footer Generation Guidelines (Antigravity Standard)

This document provides guidelines on how to programmatically inject institutional headers and footers into compiled PDF documents using Python and PyMuPDF (`fitz`).

1. Concept

Instead of relying on fragile HTML/CSS page rendering contexts that vary across layout engines, the Antigravity approach compiles standard Markdown to a clean, margin-compliant intermediate PDF. We then perform a post-processing pass using PyMuPDF to draw vector logos, document title headers, and footer blocks on every page at absolute coordinates.

2. Layout Structure

The layout bounds on a standard A4 page (595 x 842 points) are mapped as follows:

```
graph TD
    subgraph A4 Page Boundary [A4 Page Layout: 595 x 842 pt]
        subgraph Top Margin Area [Top Margin: Y=0 to Y=60]
            L_Logo["Left Logo: am.png<br/>X=48, Y=12<br/>(146.5 x 24.5
                pt)<br/>[overlay=False]"]
            M_Title["Middle Title: Document Title<br/>X=Centered in
                remaining space, Y=26"]
            R_Logo["Right Logo: Bts.png<br/>X=479.9, Y=8<br/>(67.1 x 36.8
                pt)<br/>[overlay=False]"]
        end

        subgraph Content Area [Main Document Context: Y=60 to Y=806]
            Content["HTML/CSS Layout Story Flow<br/>[borders=(36, 60, -36,
                -36)]"]
        end

        subgraph Bottom Margin Area [Bottom Margin: Y=806 to Y=842]
            L_Foot["Left Footer: ferro988 /
                your_windows_user_here<br/>X=48, Y=820"]
            M_Foot["Middle Footer: Page X of Y<br/>X=Centered, Y=820"]
            R_Foot["Right Footer: DEVOP1<br/>X=width-48-width, Y=820"]
        end
    end
```

3. High-Quality Scaled Drawing

To ensure maximum visual fidelity when zoomed, follow these steps:
1. **Load PNGs as Pixmaps**: Read the image size using PyMuPDF `fitz.Pixmap`.
2. **Compute 10% Bounds**: Scale the pixel width and height to 10% for layout dimensions in points.

3. **Draw Bounds in Background**: Insert the image using `page.insert_image(rect, filename, overlay=False)`. Setting `overlay=False` places the header pictures in the background, keeping other text on the foreground.
4. **Prevent Header Collisions**: Instantiate the markdown section with a top border margin of 60pt (e.g. `borders=(36, 60, -36, -36)`) so that layout text begins below the header logo graphics.
5. **Optimized Saving**: Always save the PDF with deflate compression, garbage collection, and linearization:

```
doc.save(pdf_file, clean=True, garbage=3, deflate=True)
```

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